In [3]:
```
import pandas as pd

data = {
    'CustomerID': range(1, 11),
    'Gender': ['Male','Male','Female','Female','Female','Female','Female','Female',
    'Age': [19,21,20,23,31,22,35,23,64,30],
    'Annual Income (k$)': [15,15,16,16,17,17,18,18,19,19],
    'Spending Score (1-100)': [39,81,6,77,40,76,6,94,3,72]
}

df=pd.DataFrame(data)
```

In [4]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              10 non-null     int64
 1   Gender                  10 non-null     str
 2   Age                     10 non-null     int64
 3   Annual Income (k$)      10 non-null     int64
 4   Spending Score (1-100)  10 non-null     int64
dtypes: int64(4), str(1)
memory usage: 532.0 bytes
```

In [5]:
```
df.head()
```

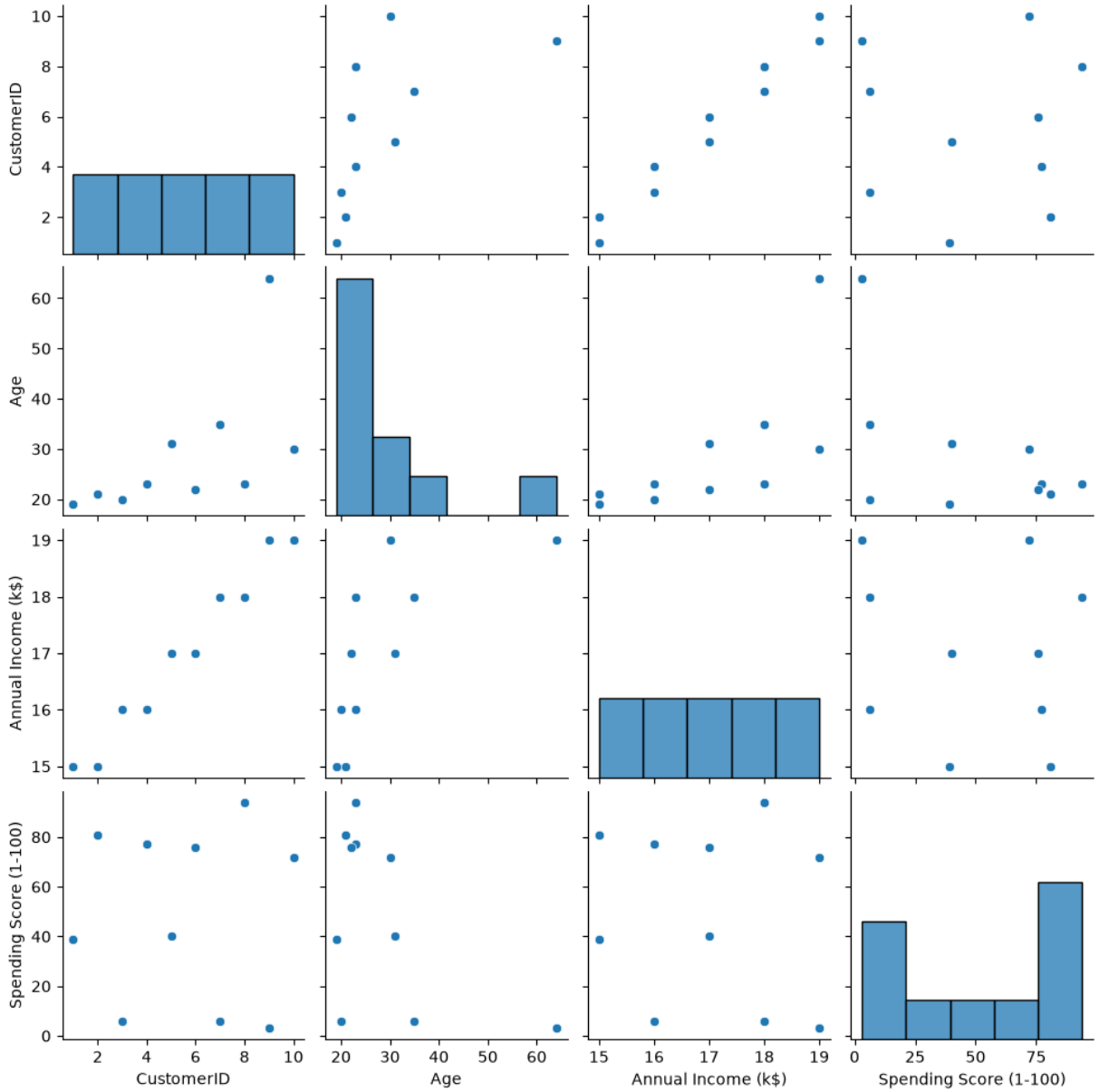
Out[5]:

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

In [6]:
```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

In [7]:
```
sns.pairplot(df)
```

Out[7]:
```
<seaborn.axisgrid.PairGrid at 0x2f2e7b59550>
```



In [8]:
```
features=df.iloc[:,[3,4]].values
```

In [9]:
```
from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)
```

Out[9]:

▾ KMeans ⓘ ?

▸ Parameters

▸ Fitted attributes

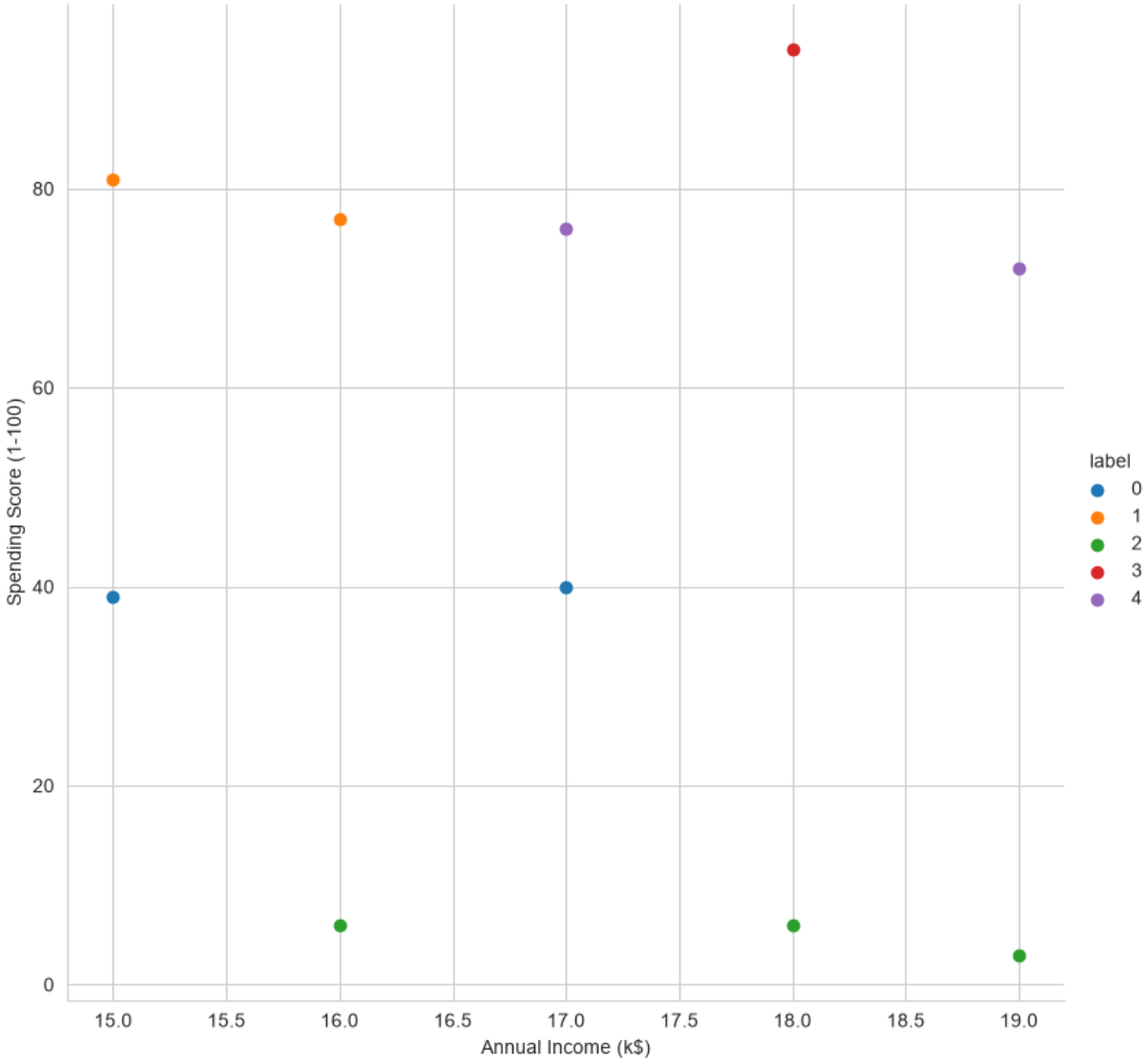
▸ 5 features

In [10]:
```
Final=df.iloc[:,[3,4]].copy()
Final['label']=model.predict(features)
Final.head()
```

Out[10]:

	Annual Income (k$)	Spending Score (1-100)	label
0	15	39	0
1	15	81	1
2	16	6	2
3	16	77	1
4	17	40	0

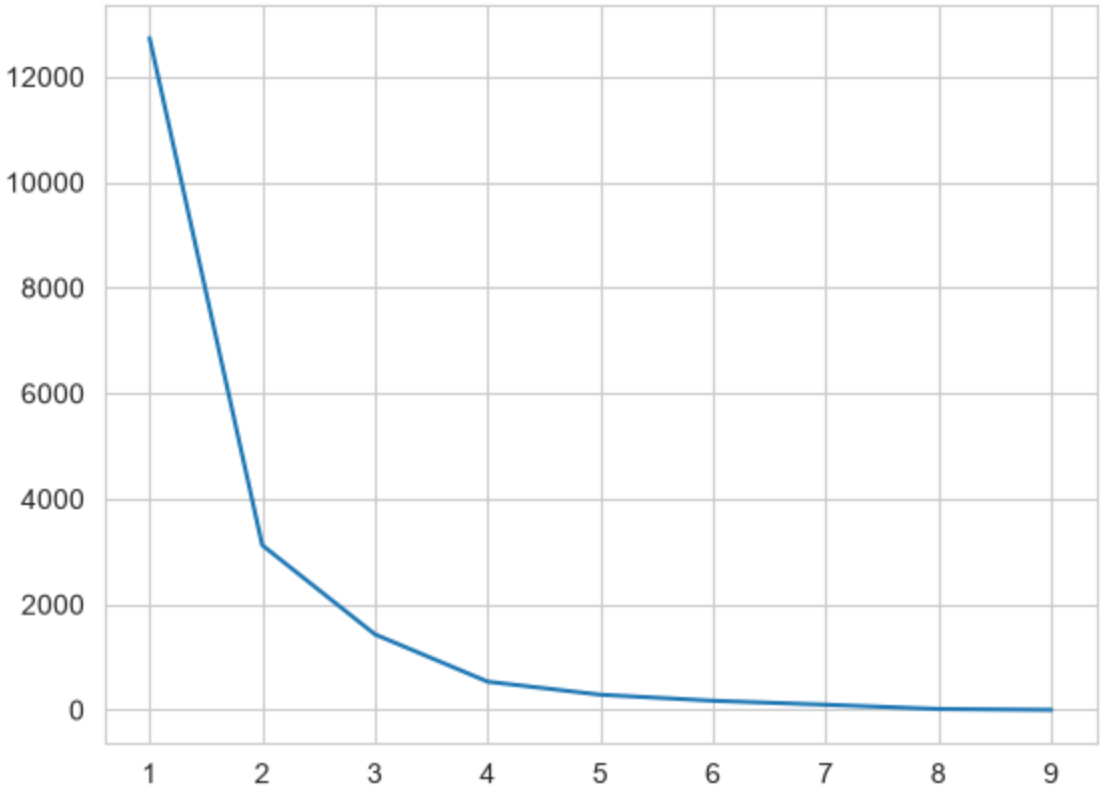
In [11]:
```
sns.set_style("whitegrid")
sns.FacetGrid(Final,hue="label",height=8) \
.map(plt.scatter,"Annual Income (k$)", "Spending Score (1-100)") \
.add_legend();
plt.show()
```



In [12]:
```
features_el=df.iloc[:,[2,3,4]].values
from sklearn.cluster import KMeans
wcss=[]
for i in range(1,10):
```

```
    model=KMeans(n_clusters=i)
    model.fit(features_el)
    wcss.append(model.inertia_)
plt.plot(range(1,10),wcss)
```

Out[12]: [<matplotlib.lines.Line2D at 0x2f2eb8692b0>]



In [ ]: